

Perishable Lot-Sizing with Period-Dependent Shelf Lives

Compact Mixed-Integer Programming Model

Indices and Sets

- $T = \{1, 2, \dots, \mathbf{T}\}$: planning periods.
- I : items.
- $i \in I$ (item), $t \in T$ (production period), $u \in T$ (consumption period), with $t \leq u$.

Perishability Windows and Arc Set

Shelf life. If item i is produced in period t and consumed in period $u \geq t$, its age is $u - t$. With period-dependent shelf life $m_{i,t} \geq 1$, a unit is usable only if

$$u \leq t + m_{i,t} - 1.$$

Consumption window from (i, t) .

$$\{u \in T \mid t \leq u \leq \min(\mathbf{T}, t + m_{i,t} - 1)\}.$$

Production periods that can cover demand (i, u) .

$$\{t \in \{1, \dots, u\} \mid u \leq t + m_{i,t} - 1\}.$$

Arc set. Decision variables $x_{i,t,u}$ are defined only when $t \leq u \leq t + m_{i,t} - 1$.

Special cases.

- $m_{i,t} = 1$: immediate consumption only ($u = t$).
- $m_{i,t} \geq \mathbf{T} - t + 1$: no effective expiry within the horizon.

Tiny example. One item, $\mathbf{T} = 7$, $(m_{i,3}, m_{i,4}, m_{i,5}) = (2, 4, 1)$:

$$\{u : (i, 3, u)\} = \{3, 4\}, \quad \{u : (i, 4, u)\} = \{4, 5, 6, 7\}, \quad \{u : (i, 5, u)\} = \{5\}.$$

Arcs $(i, 3, 5)$ or $(i, 5, 6)$ do not exist.

Parameters

$d_{i,u}$	demand of item i in period u .
c_i^{var}	variable production cost per unit.
h_i	unit holding cost per period (applied to age $u - t$).
$s_{i,t}$	setup cost for item i in period t (may be constant in t).
b_i^{var}	backlog penalty per unit (used only if backorders are allowed).
cap_t	global production capacity in period t .
$\text{cap}_{i,t}$	optional per-item capacity in period t .
$m_{i,t}$	shelf life for production of i in t .
W	optional warehouse capacity (end-of-period inventory limit).
$M_i = \sum_{u \in T} d_{i,u}$	tight big- M for setup linking.

Decision Variables

$x_{i,t,u} \in \mathbb{Z}_{\geq 0}$	quantity of i produced in t and consumed in u , only if $t \leq u \leq t + m_{i,t} - 1$.
$y_{i,t} \in \{0, 1\}$	setup indicator for i in period t .
$b_{i,u} \in \mathbb{Z}_{\geq 0}$	backlog at end of period u for i (used only in the backorder variant).

Cost Definition

Arc cost (one unit flowing from t to u).

$$C_{i,t,u} = c_i^{\text{var}} + h_i(u - t).$$

Setup cost at (i, t) : $s_{i,t}$. **Backlog penalty at (i, u) :** b_i^{var} .

Objective

$$\min \sum_{i \in I} \sum_{t \in T} \sum_{\substack{u \in T \\ t \leq u \leq t + m_{i,t} - 1}} C_{i,t,u} x_{i,t,u} + \sum_{i \in I} \sum_{t \in T} s_{i,t} y_{i,t} + \sum_{i \in I} \sum_{u \in T} b_i^{\text{var}} b_{i,u} \quad (\text{include only if backorders are allowed}).$$

Constraints

Global capacity (per production period). For $t \in \{1, \dots, T\}$,

$$\sum_{i \in I} \sum_{\substack{u \in T \\ t \leq u \leq t + m_{i,t} - 1}} x_{i,t,u} \leq \text{cap}_t.$$

Reading tip: this sums all flow out of production period t across items, but only along arcs allowed by shelf life.

Per-item capacity (optional). For $i \in I$, $t \in T$ where $\text{cap}_{i,t}$ is specified,

$$\sum_{\substack{u \in T \\ t \leq u \leq t + m_{i,t} - 1}} x_{i,t,u} \leq \text{cap}_{i,t}.$$

Special case: if $\text{cap}_{i,t}$ is not provided, omit this inequality for that (i, t) .

Setup linking (tight M_i). For $i \in I$, $t \in T$,

$$\sum_{\substack{u \in T \\ t \leq u \leq t+m_{i,t}-1}} x_{i,t,u} \leq M_i y_{i,t}.$$

Notes: $M_i = \sum_u d_{i,u}$ is tight. If $M_i = 0$ then $y_{i,t} = 0$ is forced and production of i never occurs.

Warehouse capacity (optional). Define end-of-period inventory after u as

$$\text{inv}_u = \sum_{i \in I} \sum_{t=1}^u \sum_{\substack{w \in T \\ t \leq w \leq t+m_{i,t}-1 \\ w > u}} x_{i,t,w}.$$

If W is given, enforce for $u \in \{1, \dots, T-1\}$,

$$\text{inv}_u \leq W.$$

Special case: at $u = \mathbf{T}$, $\text{inv}_{\mathbf{T}} = 0$ because there is no future consumption period.

Demand balance (no backorders). For $i \in I$, $u \in T$,

$$\sum_{\substack{t=1 \\ u \leq t+m_{i,t}-1}}^u x_{i,t,u} = d_{i,u}.$$

Reading tip: sum over $t = 1, \dots, u$ but include only those t whose shelf life at production covers consumption in u .

Demand balance (with backorders). For $u = 1$:

$$\sum_{\substack{t=1 \\ 1 \leq t+m_{i,t}-1}} x_{i,t,1} + b_{i,1} = d_{i,1}.$$

For $u \geq 2$:

$$\sum_{\substack{t=1 \\ u \leq t+m_{i,t}-1}}^u x_{i,t,u} + b_{i,u} = d_{i,u} + b_{i,u-1}.$$

Special cases and interpretation.

- If the coverage set is empty in a period u (no t satisfies $u \leq t+m_{i,t}-1$), then with backorders the entire demand must be carried in $b_{i,u}$; without backorders, the instance is infeasible.
- When $m_{i,t} = 1$, only $t = u$ contributes to the sum for that u .
- Backlog evolves as a simple “carry-forward”: demand plus prior backlog must be met by feasible production or be left as backlog.